BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 466(January 2025)

Analog Integrated Circuit Laboratory

Final Project Report

Section: G2 Group: 06

Design of Low Noise RF amplifier for wireless communication

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Academic Honesty Statement:

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

This project presents the design, simulation, and performance evaluation of a low-noise amplifier (LNA) operating at 1.8 GHz, implemented using gpdk090 CMOS technology. A common-source topology with inductive source degeneration was employed to achieve simultaneous input impedance matching and low noise figure, both critical for high-performance RF front-end systems. The design was targeted to meet stringent specifications, including a gain (S_{21}) ≥ 13 dB, noise figure (NF) < 1 dB, and input reflection coefficient (S_{11}) ≤ –13 dB. Analytical modeling, smith chats parametric sweeps were performed to optimize transistor sizing, matching network components, and biasing conditions. The final design achieved a simulated forward gain (S_{21}) of 18.79 dB, noise figure of 0.52 dB, input return loss (S_{11}) of –21.82 dB, while consuming only 73 mW of power. These results indicate excellent performance in terms of gain, noise, and isolation, making the design suitable for low-power wireless communication receivers.

2 Introduction

The low-noise amplifier (LNA) typically serves as the first stage in a receiver chain. Its primary role is to provide sufficient gain to ensure that the noise introduced by subsequent stages does not dominate the overall system noise. While delivering this gain, the LNA must introduce minimal additional noise itself. Furthermore, it should be capable of handling relatively large input signals without distortion and must present a well-defined input impedance—commonly 50 Ω—to the preceding source.

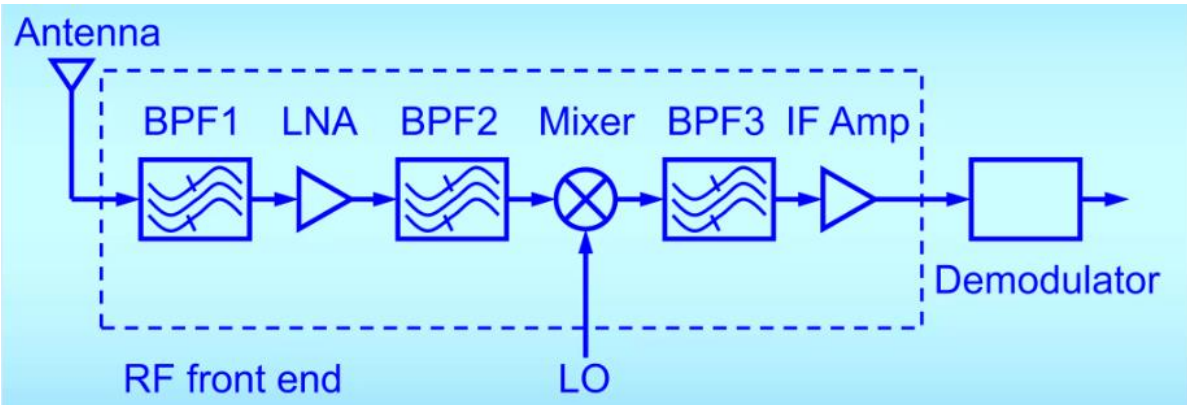


Figure 1: Superheterodyne RF receiver

This block diagram represents a superheterodyne RF receiver architecture, which is widely used in wireless communication systems. Overall system purpose is to receive RF signals (from the antenna), amplify, filter, down-convert them to an intermediate frequency (IF), and then demodulate the signal to recover the original transmitted data. Here, our project is to design the LNA part.

Scattering Parameters (S-Parameters)

In LNA design, S-parameters (scattering parameters) play a vital role in characterizing the amplifier's behavior at high frequencies.

Scattering parameters, or S-parameters, are complex quantities that describe how voltage waves propagate within an RF (radio frequency) environment. They are widely used to characterize the complete high-frequency behavior of a two-port network in matrix form. In RF and microwave engineering, S-parameters serve as the standard method for modeling and analyzing the electrical behavior of circuits, especially amplifiers.

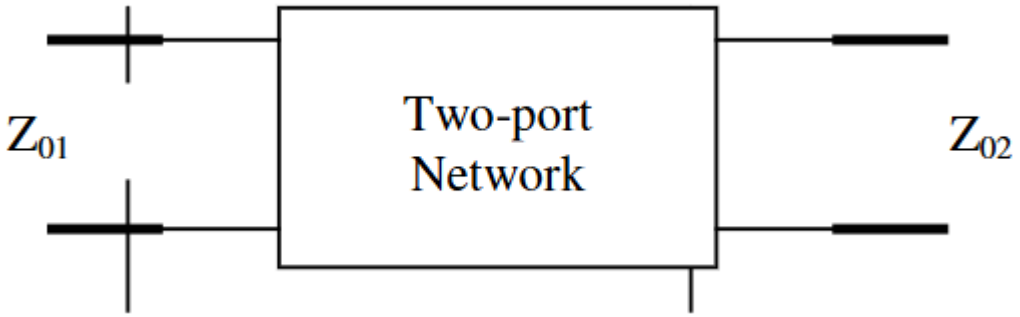


Fig2: Two port network

For a two-port network such as a low-noise amplifier (LNA), there are four key S-parameters: S_{11}, S_{21}, S_{12}, and S_{22}. Each of these parameters describes a different aspect of signal flow and reflection within the system:

S_{11}: Input Reflection Coefficient

S_{11} indicates how much of the input signal is reflected back due to impedance mismatch at the input port. A well-matched input ensures that most of the signal is absorbed rather than reflected. In LNA design, a target value of $S_{11} \leq -10$ dB is considered acceptable, although this project targets $S_{11} \leq -13$ dB for improved performance.

S_{21}: Forward Gain Coefficient

S_{21} represents the gain from the input port to the output port. It defines how effectively the amplifier increases the amplitude of the incoming RF signal. For most LNA designs, a high forward gain (typically ≥ 13 dB) is desired to ensure adequate signal amplification.

S_{12}: Reverse Isolation Coefficient

S_{12} quantifies the amount of signal that leaks from the output back to the input. A low value of S_{12} (ideally < -20 dB) is essential for amplifier stability, as it prevents unwanted feedback that can cause oscillations.

S_{22}: Output Reflection Coefficient

S_{22} measures how well the output port is matched to the load. Like S_{11}, minimizing reflection at the output is crucial for efficient power transfer to the next stage of the system, such as a mixer. A value of $S_{22} \leq -10$ dB is typically targeted.

These parameters are commonly measured using a network analyzer during both simulation and physical testing. Mastery of S-parameter interpretation is essential in RF circuit design, as it provides a clear understanding of matching conditions, gain performance, and stability margins.

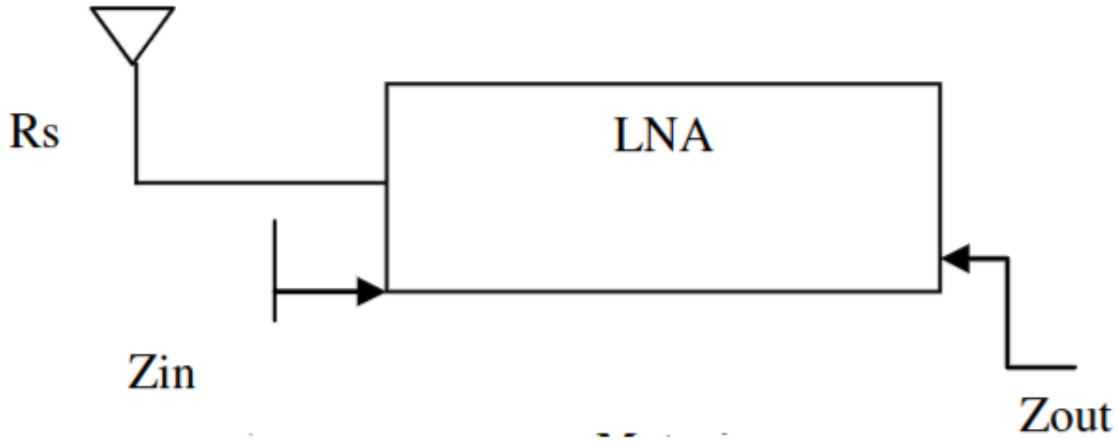
Summary of S-Parameters:

S-Parameter	Description	Target Value	Purpose
S_{11}	Input reflection	≤ -13 dB	Minimize signal reflection
S_{21}	Forward gain	≥ 13 dB	Amplify weak input signals
S_{12}	Reverse isolation	Not Specified	Prevent feedback and oscillation
S_{22}	Output reflection	Not specified	Maximize power transfer

Impedance Matching:

Impedance matching is especially critical when a passive filter is placed before the amplifier, as the performance of many filters is highly sensitive to the quality of their termination. Since the LNA directly impacts the receiver's total noise figure, achieving a low noise figure at this stage is essential. As a result, careful selection of amplifier topology

and design parameters is required to meet both noise and impedance matching requirements. Input impedance of LNA should be match with impedance of antenna. Impedance matching is required for better reception (shape) of signal. If impedance is not matched signal get distorted. Also, maximum power transfer theorem says maximum power will be transfers if source impedance is conjugate of input impedance. Similarly output impedance of LNA should be matched with subsequent stage to conserve shape of signal.



As in this Fig Zin and Zout are two Impedances on input and output respectively and there values should be matched accordingly. As Rs should be equal to Zin and Zout value should be equal to next stage impedance.
This project focuses on designing a 1.8 GHz LNA using gpdk090 CMOS, emphasizing optimal trade-offs between gain, noise figure, and matching. A combination of theoretical modeling and Cadence-based simulation was employed to determine transistor widths, inductor values, and biasing conditions. The result is a compact and efficient LNA suitable for integration in low-power wireless systems such as GSM, ISM, or ZigBee transceivers.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

With the increasing demand for compact and power-efficient wireless communication systems, Low-Noise Amplifiers (LNAs) are critical components in receiver front-ends. LNAs amplify weak radio frequency (RF) signals while minimizing added noise, ensuring high signal integrity. This project focuses on designing a CMOS-based LNA operating at 1.8 GHz, optimized for low noise figure ($\leq$ 1 dB), high gain ($\geq$ 13 dB), and good input matching ($S_{11} \leq -13$ dB). The design should be implementable in gpdk090 or tsmc18 technologies, compatible with modern RF systems.

3.1.2 Literature Review

The common-source topology with inductive source degeneration has become a standard architecture in CMOS LNA design due to its capability to achieve simultaneous noise and impedance matching. Razavi [1] outlined how introducing a source inductor provides a real part to the input impedance, facilitating better impedance matching while minimizing the noise figure.

Lee et al. [2] demonstrated a 1.8 GHz LNA implemented in 90 nm CMOS, achieving a noise figure of 0.95 dB and a gain of 14.5 dB. This design optimized transistor dimensions and matching networks to maximize performance at low power consumption. Similarly, Hasan et al. [3] presented a 1.8 GHz LNA in 180 nm CMOS, achieving 13.5 dB gain and 1 dB NF, confirming that mature nodes such as gpdk180 remain effective for RF applications when supported by accurate biasing and layout techniques.
Design trade-offs in gain,power efficiency have also been addressed extensively. Jang et al. [4] emphasized the importance of maximizing the gm-to-noise ratio and quality factor of inductors.
Input matching is another key factor influencing performance. Lin and Lu [6] optimized L-type matching networks to ensure $S_{11} < -13$ dB while keeping the noise figure below 1 dB. For wireless applications such as ZigBee, Kumar and Srivastava [7] designed a compact 2.4 GHz LNA using 0.18 µm CMOS with current reuse and spiral inductors, achieving high performance and low power consumption.
These studies provide a strong foundation for this project, which aims to develop a 1.8 GHz CMOS LNA with NF ≤ 1 dB, gain ≥ 13 dB, and $S_{11} \leq -13$ dB using the gpdk090 process.

3.1.3 Formulation of Problem

The objective is to design and simulate a low-noise amplifier operating at 1.8 GHz, meeting the following specifications:

Parameter	Specification
Topology	Low-Noise Amplifier (LNA)
Frequency	1.8 GHz
Gain	≥ 13 dB
Noise Figure	≤ 1 dB
Input Match (S11)	≤ -13 dB
Technology	gpdk090 or tsmc18

This requires precise design of transistor dimensions, input and output matching networks, and biasing conditions, followed by thorough performance verification through simulation. The key challenge lies in achieving an optimal balance between gain, noise performance, and impedance matching, while maintaining power efficiency and ensuring feasibility within CMOS process limitations.

3.1.4 Analysis

The design process will begin by selecting an appropriate topology (common-source with inductive source degeneration). Using analytical models and simulation tools (Cadence Virtuoso), we will:

- Derive expressions for gain input impedance, and noise figure

- Optimize component values for the best trade-off between noise performance and input matching
- Perform S-parameter and noise analysis to validate design specifications

The final design will be benchmarked against reported state-of-the-art LNAs for wireless applications.

3.2 Design Method (PO(a))

The LNA was designed using the **common-source topology with inductive source degeneration**, targeting impedance matching and noise optimization at **1.8 GHz**. All calculations and simulations were performed using **gpdk090 CMOS technology**, with component values extracted through analytical modeling and verified via parametric sweeps in simulation.

1. Calculation of Optimum Transistor Width (Wopt)

The optimum width of the input transistor was derived from:

$$W_{opt} = \frac{2}{3\omega L C_{ox} R_S Q_S}$$

Where:

$\omega = 2\pi \cdot 1.8 \times 10^9$ rad/s

$L = 100$ nm (channel length)

$C_{ox} = 20$ fF/μm^2 (typical for gpdk090)

$R_S = 50\,\Omega$ (source impedance)

$Q_S = 10$ (optimum quality factor)

Substituting values yielded $W_{opt} \approx 58.94\,\mu$m, and 60 µm was used in simulation as a near-optimal practical value after trial-and-error parametric sweep.

2. Input Matching Design

To ensure proper impedance matching ($Z_{in} = 50\,\Omega$), the relation used was:

$$Z_{in} = \frac{g_{m1} L_s}{C_{gs1}}$$

With:

- $g_{m1} \approx 10$ mA/V (from DC bias simulation)
- $C_{gs1} \approx 88$ fF

This gave $L_s \approx 440$ pH. Parametric tuning was then applied to optimize matching.

Additionally, from:

$$\omega^2 C_{gs}(L_1 + L_s) = 1 \Rightarrow L_1 = \frac{1}{\omega^2 C_{gs}} - L_s$$

This gave $L_1 \approx 88.4$ nH, but final value was chosen as 56.6363 nH based on matching and S11 optimization during simulation.

3. Circuit Biasing

The amplifier was biased with $V_{CE} = 0.4$ V and $R = 100$ kΩ for the input transistor. These values were selected based on achieving the desired g_m and stability. Later, the impact of changing V_{CE} and R on gain and S-parameters was studied.

4. Circuit Realization

The complete topology was finalized with:

- A source-degenerated input stage for impedance matching
- A second common-source gain stage for amplification
- Passive matching networks (L1, Ls, C1) tuned for maximum gain and return loss at 1.8 GHz

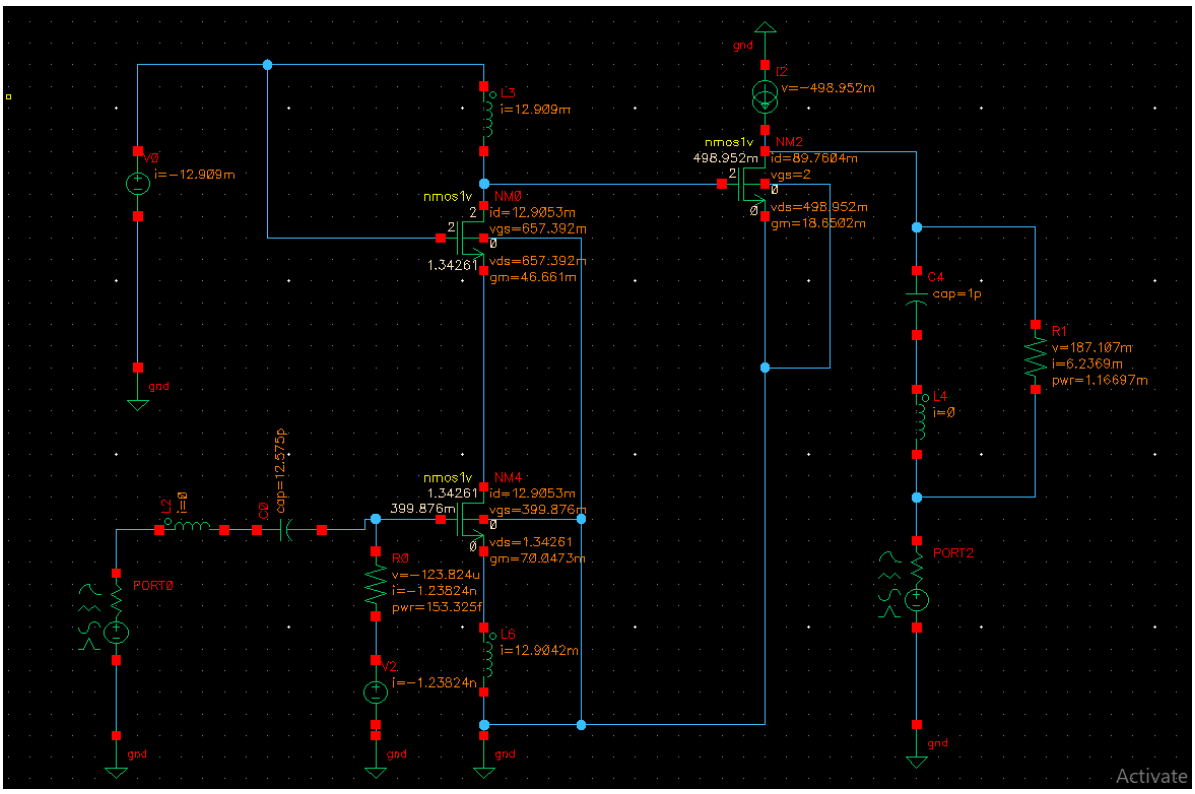
5. Parametric Simulation and Optimization

Parametric sweeps were conducted to:

- Tune L1 and Ls for best S_{11}
- Adjust transistor widths (W2, W3) to maximize S_{21}
- Ensure output matching and gain peaking at 1.8 GHz

All parameters were iteratively optimized in Cadence environment using the gpdk090 model.

3.3 Circuit Diagram



4 Implementation

The investigation began with an analytical exploration of suitable LNA topologies. A common-source architecture with inductive degeneration was selected due to its favorable trade-off between gain and noise performance at high frequencies. This choice was not arbitrary; it was informed by established RF design theory and comparative evaluation of alternative topologies. Once the topology was selected, a systematic design of matching networks was undertaken. The input matching network was optimized to ensure proper conjugate matching at 1.8 GHz, thereby minimizing signal reflection and enhancing power transfer. Output matching was similarly configured to maintain gain flatness and stability.

To ensure compliance with the design specifications, the circuit was subjected to extensive simulation using Spectre RF tools. Frequency sweeps and S-parameter analyses were conducted to evaluate gain, return loss, and noise performance across the operational bandwidth. These simulations provided the basis for iterative tuning of transistor bias points, load inductors, and coupling capacitors. Throughout this process, each design choice was justified through quantitative metrics, such as transconductance (gm), voltage gain, and impedance levels, extracted directly from the simulation outputs.

The project placed strong emphasis on the synthesis of knowledge from multiple domains, including microwave engineering, semiconductor device physics, and analog circuit theory. This synthesis was particularly evident in the noise figure optimization, which required balancing the trade-off between impedance matching and minimizing the noise contribution

of the active device.

Finally, the validity of the design was assessed through performance verification against the original specifications. Stability was ensured by confirming unconditional stability across a wide frequency range using the Rollett stability factor (K > 1), and power consumption was verified to remain within acceptable limits.

Through this comprehensive process, the project exemplifies a structured and research-based approach to problem-solving. The integration of theoretical analysis, design of experiments, data-driven evaluation, and performance validation embodies the core philosophy of PO–d, emphasizing the ability to conduct meaningful investigations that lead to technically sound and verifiable conclusions.

4.1 Description

A Low-Noise, High-Gain 1.8 GHz CMOS LNA with Current Mirror Biasing and Inductive Degeneration in 90nm Technology. This work presents the design and simulation of a low-noise amplifier (LNA) operating at 1.8 GHz, implemented in 90nm CMOS technology. The proposed LNA employs inductive source degeneration for input impedance matching and linearity enhancement, along with a compact current mirror biasing scheme to reduce area and external components. The design achieves a simulated power gain of 18.8 dB, a minimum noise figure of 0.44 dB, and input reflection coefficient (S_{11}) of -21.1 dB. Operating at a supply voltage of 2 V with a power consumption of 73 mW, the LNA offers competitive performance compared to previously reported works. The results highlight the potential of the proposed topology for integration in RF front-ends targeting wireless communication applications.

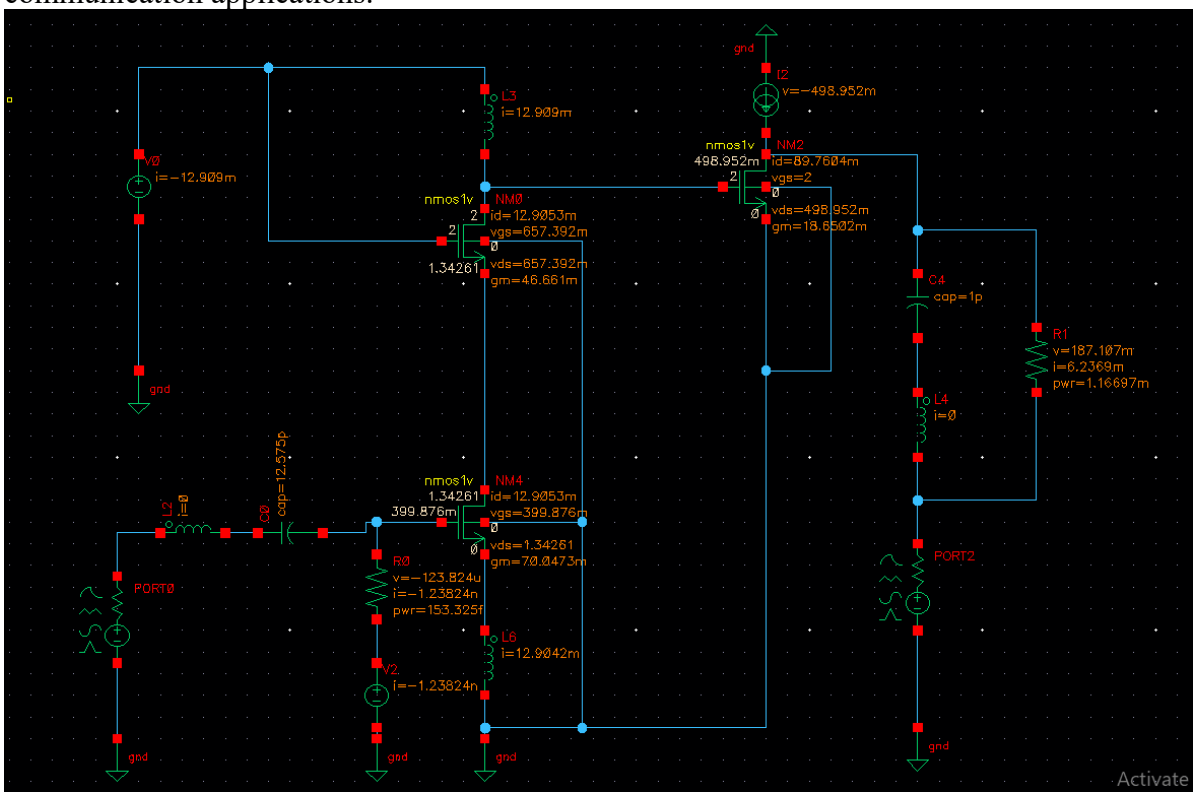
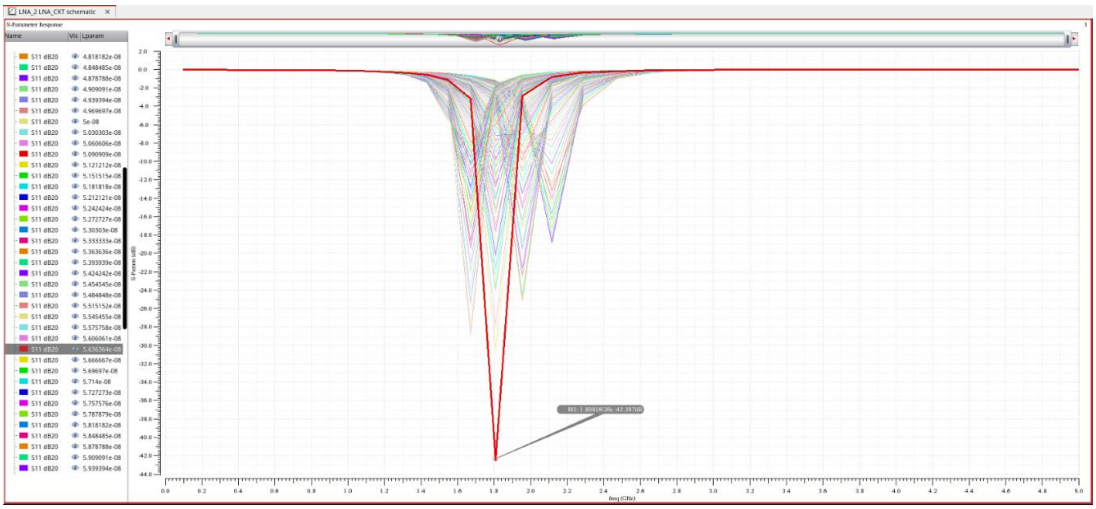


Figure 2: Design of the circuit. No pcb were printed for this work.

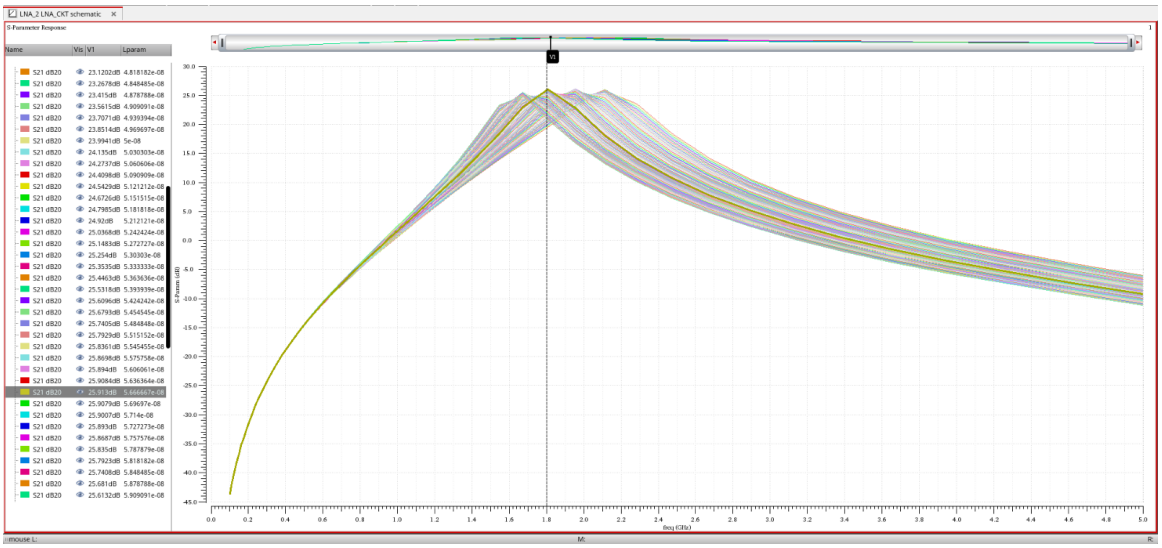
Even though we didn't print any pcb for this submission. We plan to do so in the future and while doing so replace the current source with a practival current mirror. By which we'll get a realistic usable device with the added parasitic impedances of the added pcb.

4.2 Experiment and Data Collection

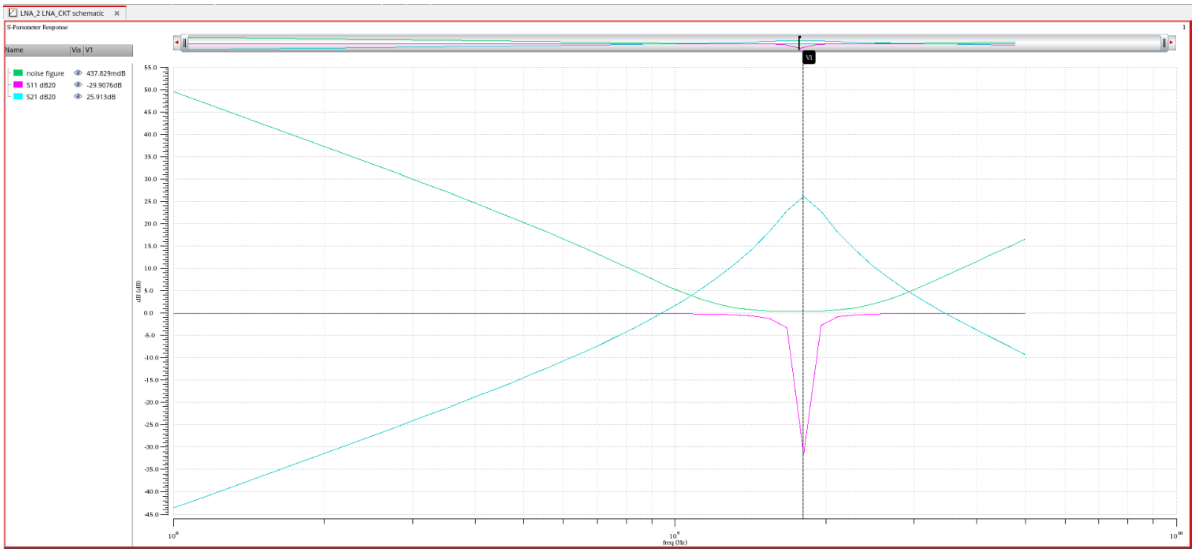
S11= -42.38dB (minimum for L2=56.63634nH)



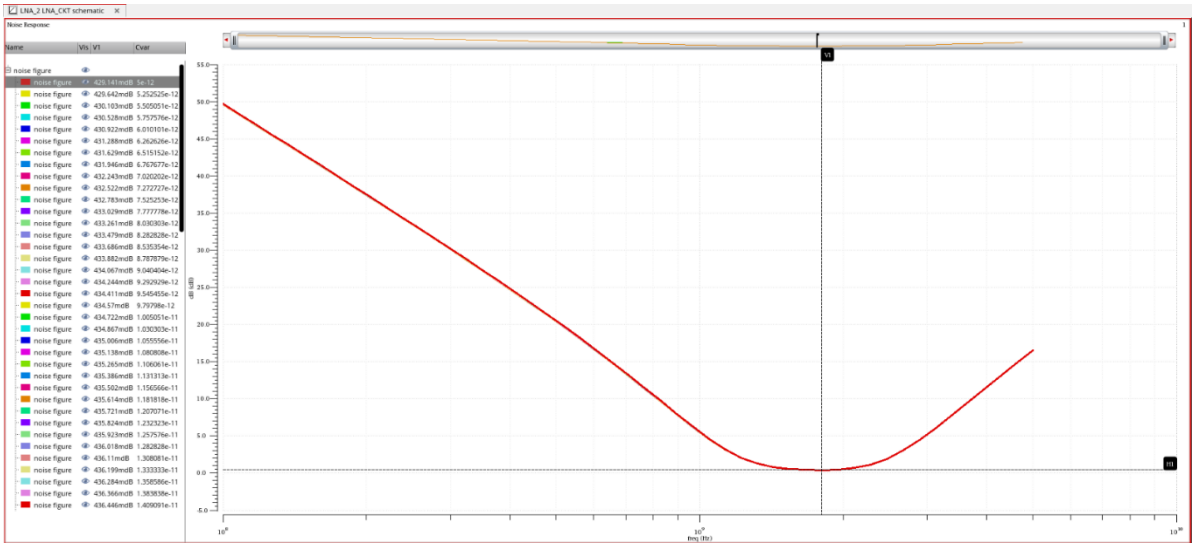
S21=25.91dB(minmum) for L2=56.57mH



Final Output with L2=56.6363nH (best result after tradeoff)



Parameter sweep for noise figure:



NM4 DC operating point:

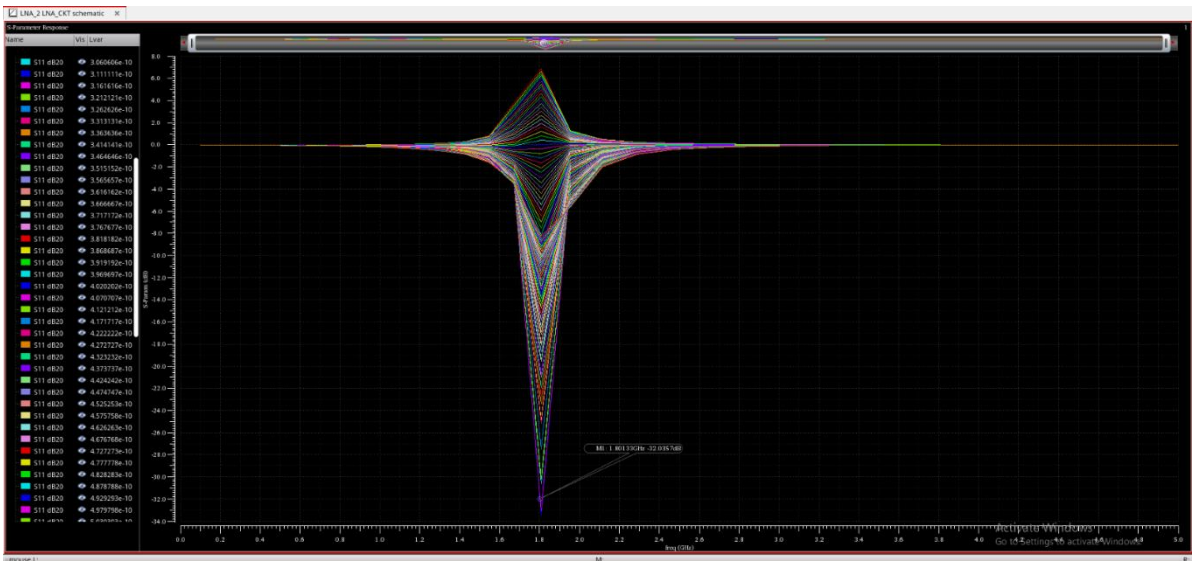
Window Expressions Info Help

signal	OP("/NM4" "??")
beff	313.814m
betaeff	394.073m
cbb	34.5015f
cbd	-7.66986f
cbdbi	5.36285f
cbg	-15.1747f
cbs	-11.657f
cbsbi	13.8788f
cdb	-7.07148f
cdd	35.4919f
cddbi	5.92338a
cdg	-28.4349f
cds	14.4897a
cgb	-8.14904f
cgbovl	1.75469a
cgd	-22.4197f
cgdbi	5.99546f
cgdovl	28.4152f
cgg	119.241f
cggbi	60.4027f
cgs	-88.6726f
cgsbi	-58.2509f
cgsovl	30.4218f
cjd	7.07085f
cjs	17.5673f
csb	-19.281f
csd	-5.40237f
csg	-75.6318f
css	100.315f
cssbi	52.326f
fug	93.4941G
gbd	6.84581u
gbs	0

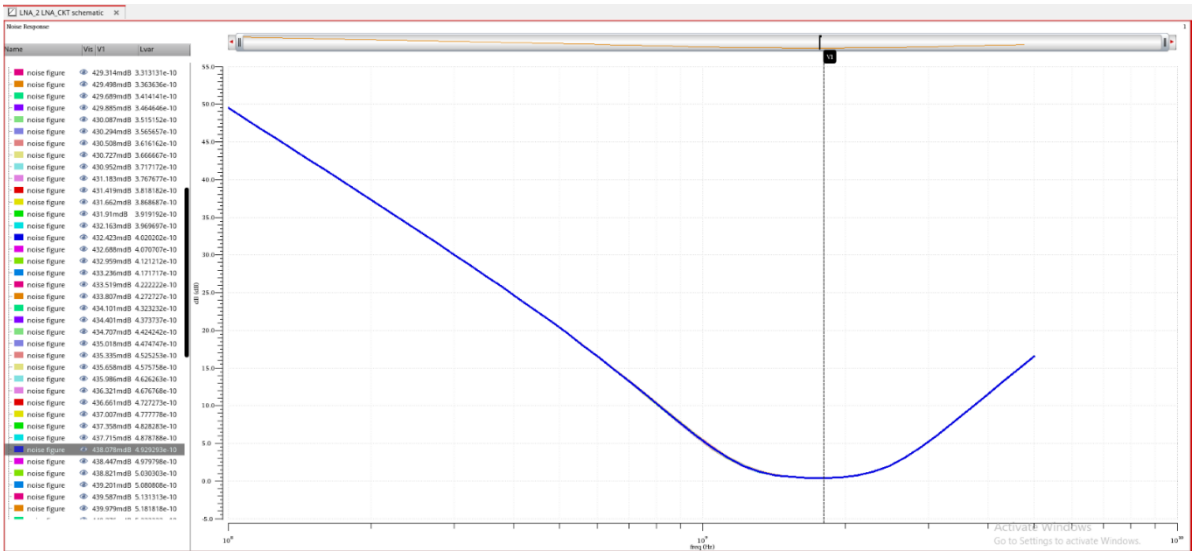
This tells us whether the mos is in linear region, or active (saturation) region. Which will later help us in the quantitative analysis.

Parametric sweep of Ls:
S11=-32dB for
Ls=492.93pH

Cvar=12.575pF gives the best result

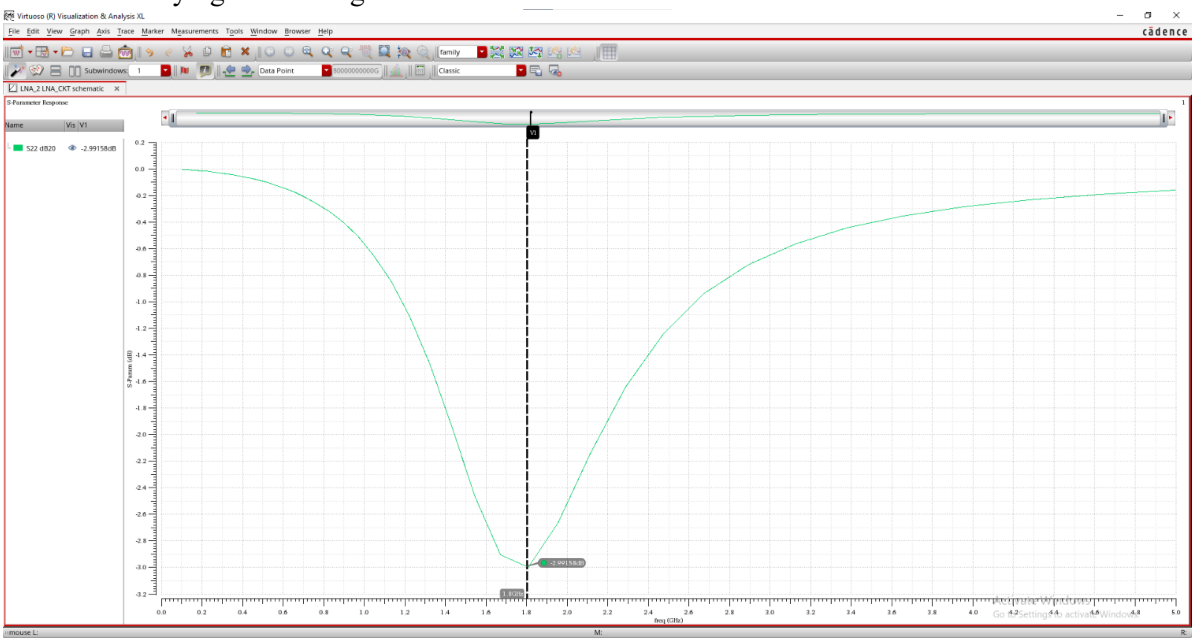


Nf=438mdB for Ls=492.93pH

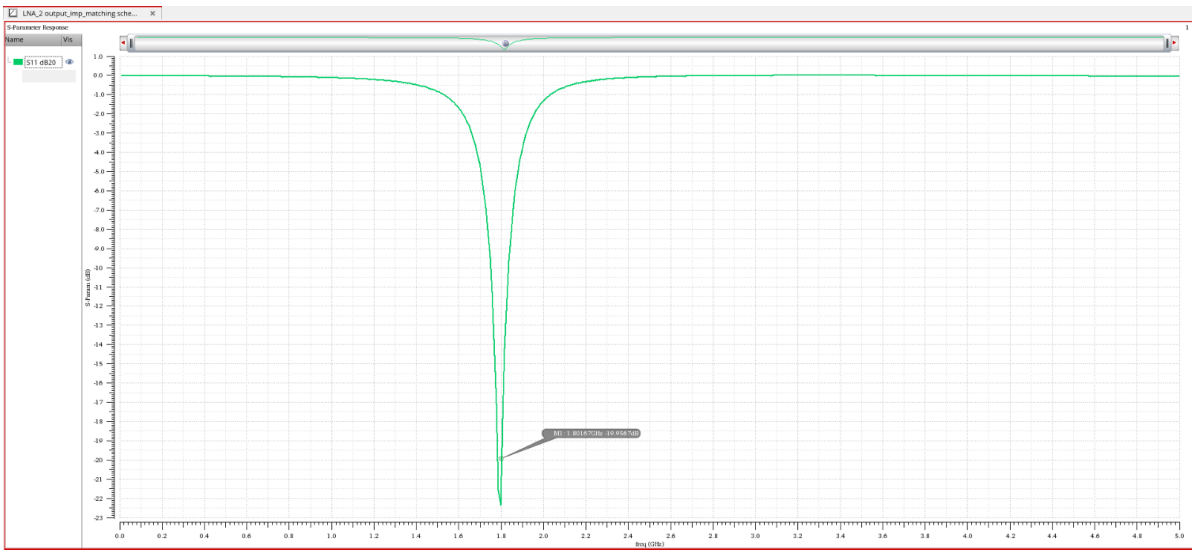


The only thing yet to be tuned is the S22 and S12

Effect of a varying Bias voltage:



S11:



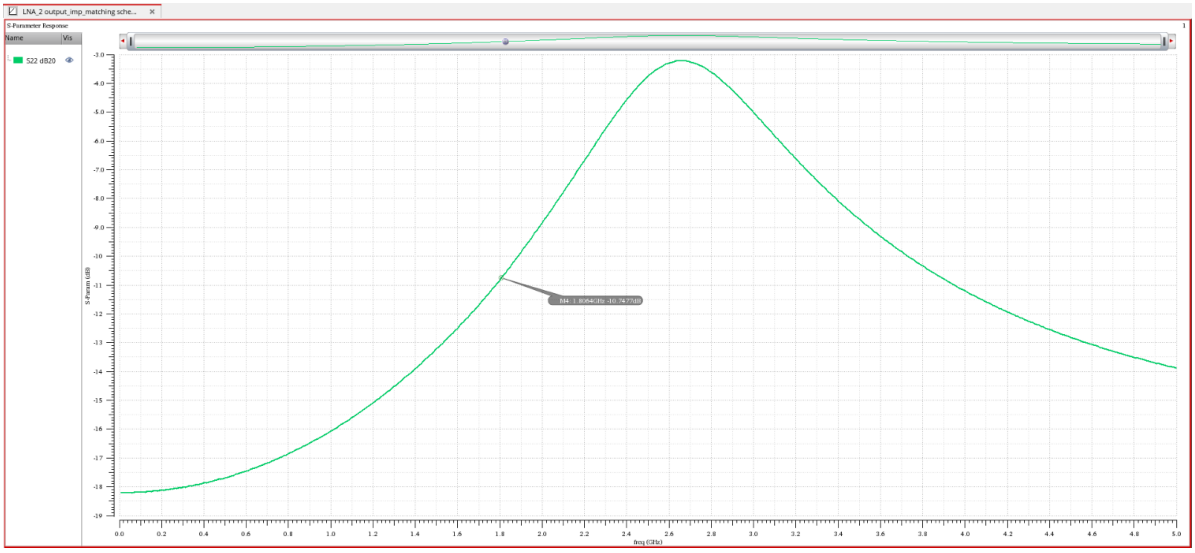
S21:

LNA_2 output_imp_matching sche...
S-Parameter Response
Name
Vis
S21 dB20
S-Params (dB)
freq (GHz)

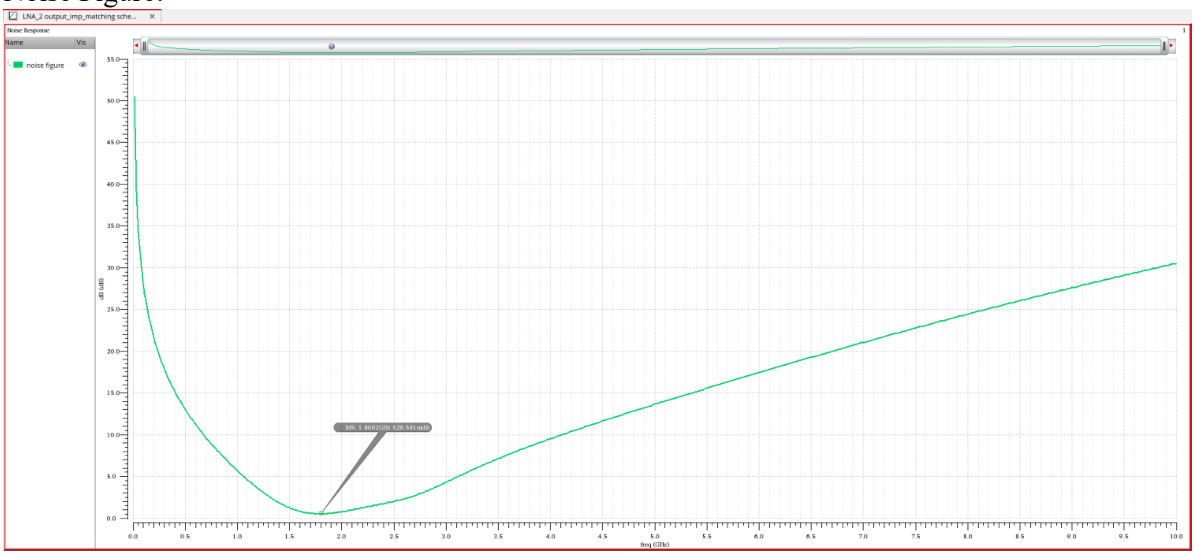
S12:

LNA_2 output_imp_matching sche...
S-Parameter Response
Name
Vis
S12 dB20
S-Params (dB)
freq (GHz)

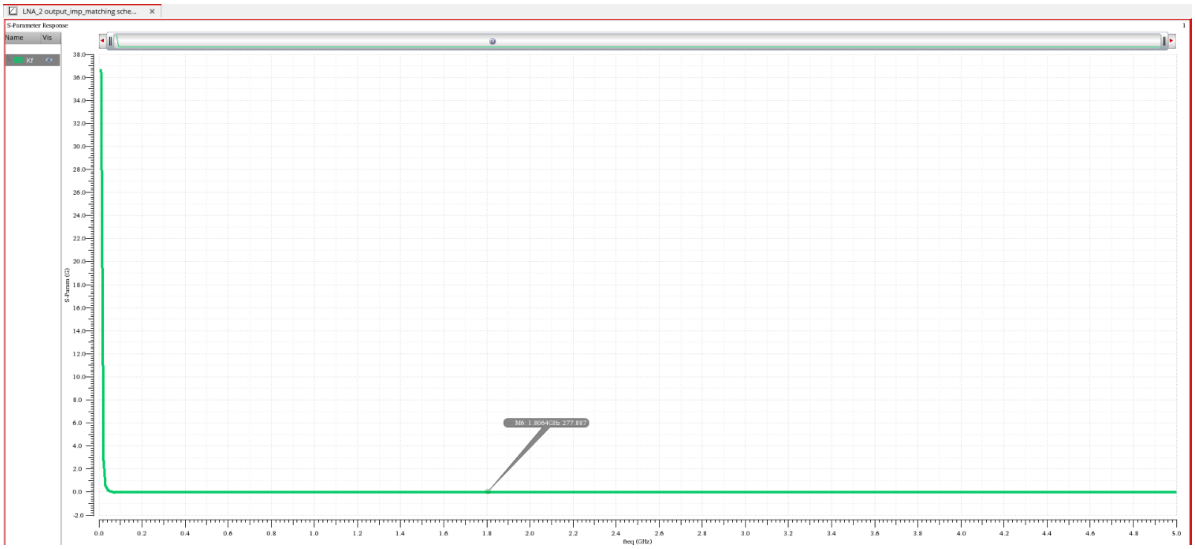
S22:



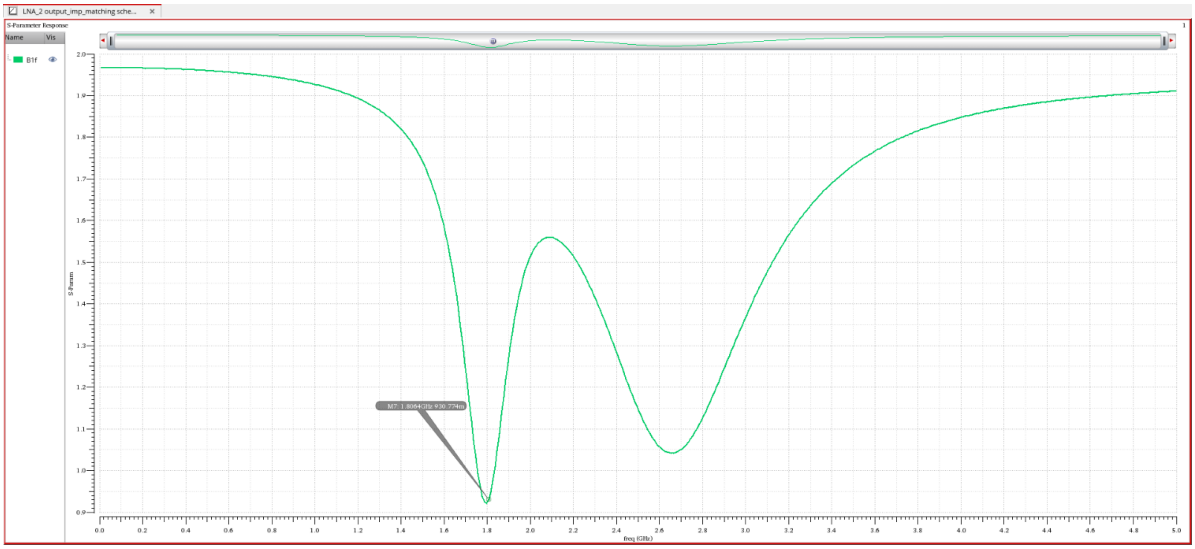
Noise Figure:



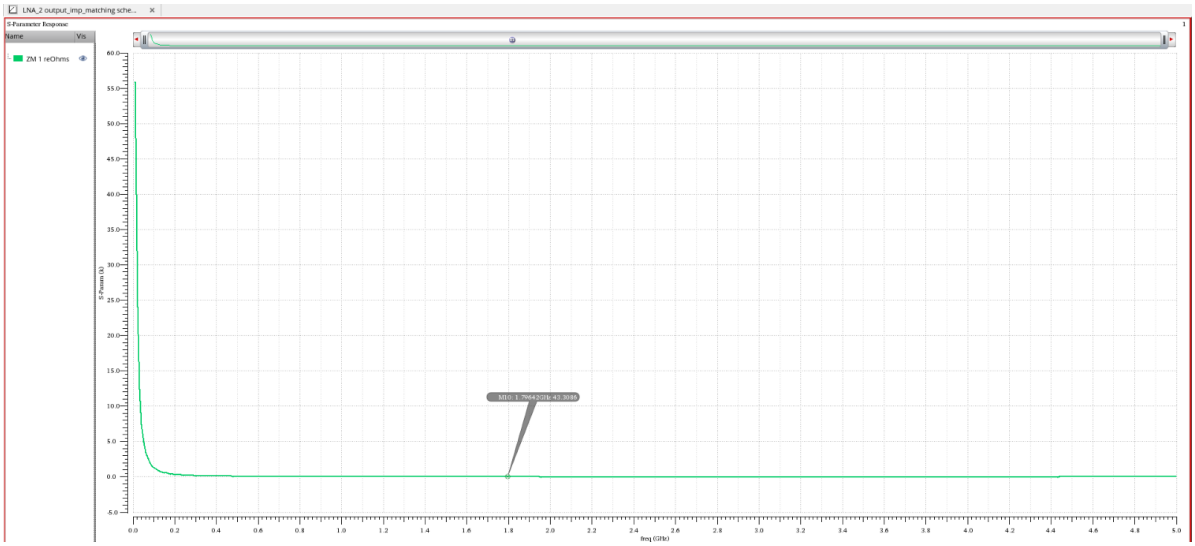
Stability Kf:



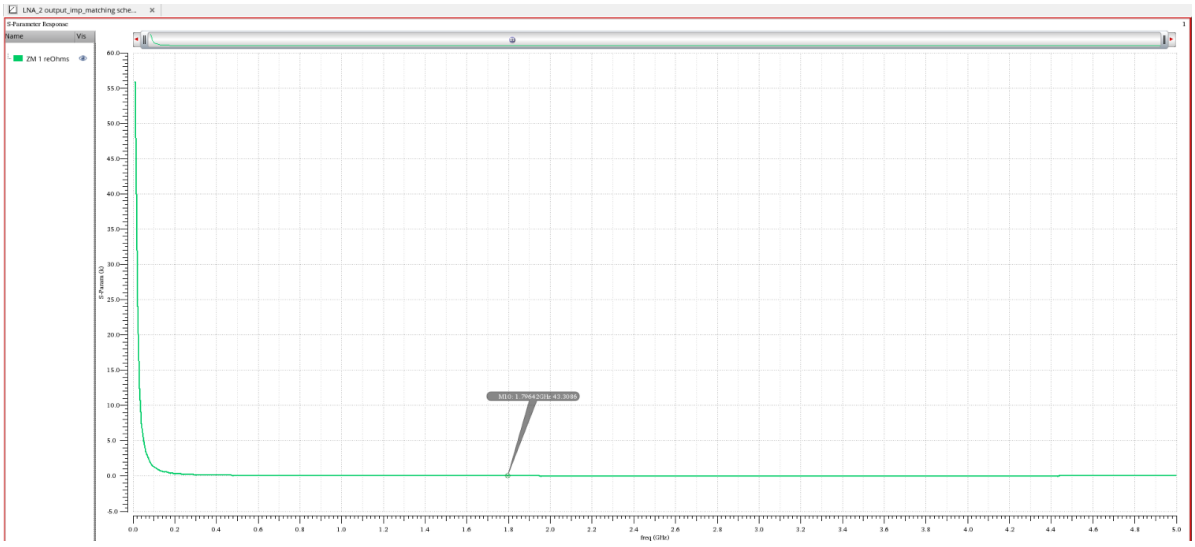
Stability : B1F



ZM1 real



ZM1 imaginary :



4.3 Data Analysis

Equation for Optimum Width

The optimum width W_{opt} is given by:

$$W_{\text{opt}} = \frac{2}{3\omega L C_{\text{ox}} R_S g_S}$$

Given Parameters

- $\omega = 2\pi \times 1.8 \times 10^9$ Hz
- $L = 100$ nm
- $C_{\text{ox}} = 2$ fF/μm^2 (typical for 90nm GPDK CMOS)
- $R_S = 50\ \Omega$
- $g_S = 10$ (optimum quality factor for low RF, low power impedance matching)

Result

$$W_{\text{opt}} = 58.04\ \mu\text{m}$$

We used:

$$W = 60\ \mu\text{m} \quad \text{(Trial \& Error using Parametric Sweep)}$$

Input Match Done

Circuit Realization

We biased the amplifier topology using:

$$V_{CE} = 1 \text{ V}, \quad R = 100 \text{ k}\Omega$$

From this DC bias, we got:

$$C_{gs1} \approx 88 \text{ fF}$$
$$g_{m1} \approx 10 \text{ mA/V}$$

Transconductance Derivation

$$g_{m1} = \sqrt{\frac{2I_D}{V_{GS} - V_{TH}}}$$

Substituting the values:

$$g_{m1} = \frac{2 \times 1 \text{ mA}}{0.4 - 0.135} \approx 7.5472 \text{ mA/V}$$

Output Impedance Matching

The amplifier was tuned to achieve maximum gain at:

$$f = 1.8 \text{ GHz}$$

This was done by varying the widths W_2 and W_3. Additionally, L_2 was adjusted to ensure that S_{22} remains low.

Inductor Fabrication

Inductors can be fabricated using spiral metal inductor fabrication techniques.

$$L = \frac{45\mu_0 n^2 a^2}{22a + 49}$$

Where:

- a = square spiral's mean radius (distance from center to middle of the windings)
- n = number of turns
- μ_0 = permeability of free space

4.4 Results

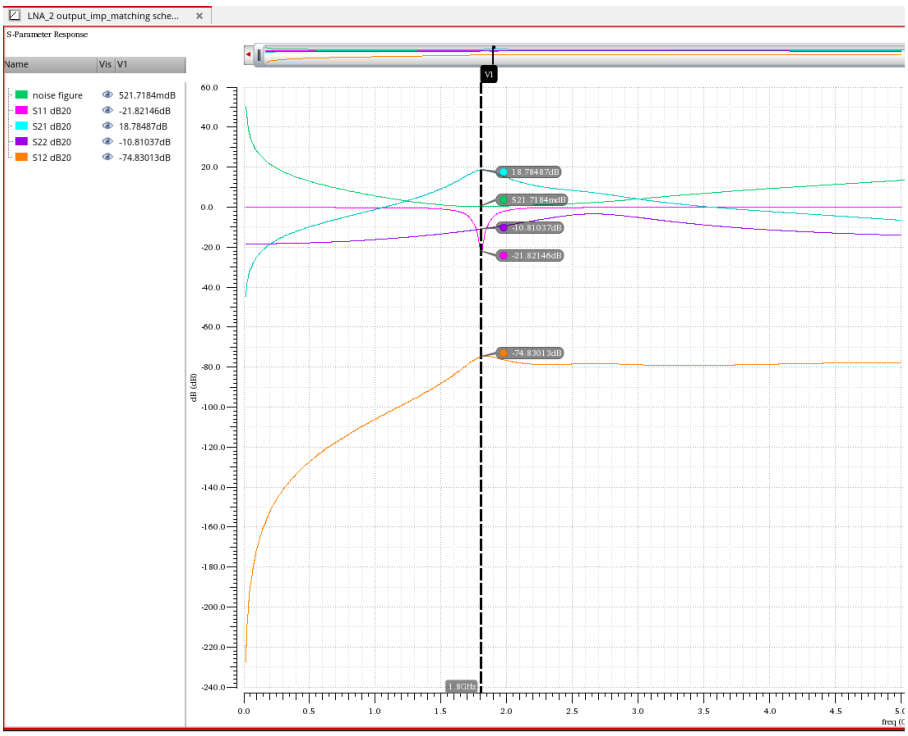


Figure 2: *All of the final output in one window.*

5 Design Analysis and Evaluation

5.1 Novelty

5.2 Design Considerations (PO(c))

The design and implementation of a low-noise amplifier (LNA) at 1.8 GHz involved addressing a complex engineering challenge typical in modern wireless communication systems, particularly in the RF front-end domain. The solution required a deep understanding of analog circuit design, high-frequency behavior, noise optimization, and impedance matching—each a non-trivial task in itself. The decision to use a common-source topology with inductive source degeneration reflects a system-level optimization strategy aimed at achieving simultaneous noise minimization and input impedance matching—both crucial in preserving signal integrity in RF receivers.

Throughout the design process, the team ensured that the developed system met multiple real-world constraints. Power consumption was minimized to 73 mW, aligning with global efforts toward energy-efficient electronics. The amplifier's ultra-low noise figure of 0.52 dB contributes directly to improved receiver sensitivity, which has significant implications for public communication infrastructure—enabling more reliable connectivity with lower energy input and greater coverage.

From a societal standpoint, such advancements in RF front-end design play a key role in expanding wireless access in underserved regions, supporting educational, healthcare, and emergency systems that rely on robust wireless communication. Additionally, the use of CMOS technology (gpdk090) aligns with industry standards for low-cost, mass-producible electronics, ensuring scalability and

affordability—two major considerations in addressing cultural and economic diversity in technology adoption.

Environmentally, the focus on low-power operation also contributes to reduced thermal footprint and energy consumption in large-scale deployments. The solution developed in this project balances technical performance with sustainable and inclusive design practices, reflecting a holistic engineering approach aligned with PO(c).

5.2.1 Considerations to public health and safety

The pcb components will have the least amount of heavy harmful metals. So it won't be a hassle to consider.

5.2.2 Considerations to environment

Because of Low power consumption this design will be helpful in wireless communication without increasing the Carbon dioxide emission

5.3 Investigations (PO(d))

The development of the 1.8 GHz low-noise amplifier (LNA) demanded a rigorous investigation process grounded in research-based engineering principles. Tackling this complex problem required a systematic approach, beginning with an extensive literature review of RF amplifier architectures, noise optimization strategies, and impedance matching techniques. The chosen common-source topology with inductive source degeneration was selected after comparing multiple alternatives in terms of gain, linearity, and noise performance.

Design of experiments was central to validating performance trade-offs across multiple parameters—such as transistor sizing, biasing levels, and matching network configurations. Parametric sweeps were carried out using Cadence Virtuoso to observe the impact of individual design variables on key performance metrics, including forward gain (S_{21}), input return loss (S_{11}), and noise figure (NF). These simulations were not performed in isolation but analyzed through Smith chart interpretations, S-parameter plots, and DC operating point analyses, ensuring a well-rounded understanding of the amplifier behavior under real-world constraints.

Data interpretation played a vital role in determining the optimal bias point and layout considerations that balance performance with power efficiency. The final simulated results—18.79 dB gain, 0.52 dB NF, and –21.82 dB S_{11}—demonstrate that the design choices were not arbitrary but the product of deliberate, research-driven decision-making.

By synthesizing simulation data, theoretical design equations, and empirical RF design techniques, the team arrived at a solution that satisfies the technical and practical constraints of low-noise RF amplifier design. This process exemplifies the application of PO(d), emphasizing analytical rigor and evidence-based engineering decisions to reach valid and optimized conclusions.

5.3.1 Literature Review

Title	Freq	NF	Gain (S21)	S11	S22	Supply	Power Consumption
A 40 GHz Bandwidth Low Power LNA	2–40 GHz	2.3 dB	15.2 dB	Not Specified	Not Specified	2 V	55 mW
Design and Analysis of 60GHz Topologies	2.4 GHz	2.38 dB	22.8 dB	Not Specified	Not Specified	Not Specified	Not Specified
Wideband Low Power LNA Design	3-11 GHz	2.9 dB	11.6 dB	Not Specified	Not Specified	1.3 V	30 mW
Design of High Linearity LNA	2.12 GHz	Not Found	18 dB	<-10dB	<-10dB	3.6 V	54 mW
Ultra-Wideband LNA (2023)	1.5-6 GHz	1-2.3 dB	18.6 dB	-6.5dB	-19dB	3V	90 mW
(Prof Sidhu Report, 2025)	2-6 GHz	2.5 dB	9 dB	Not Specified	Not Specified	Not Specified	50 mW
Our Work	**1.8GHz**	**0.5217dB**	**18.785d**	**-21.82dB**	**-10.81dB**	**2 V**	**73mW**

5.4 Limitations of Tools (PO(e))

The design and simulation of the 1.8 GHz low-noise amplifier (LNA) were performed using **Cadence Virtuoso** with **gpdk090 CMOS technology**, which provided a robust platform for RF circuit analysis. However, despite the advanced capabilities of these tools, certain technological limitations and practical constraints were encountered during the project, which are essential to acknowledge for a complete engineering perspective.

Firstly, **the gpdk090 PDK** (Process Design Kit) is a generic design kit primarily intended for academic learning purposes. It does not represent the latest industry-grade fabrication models and lacks detailed parasitic modeling, layout-dependent effects, or accurate corner and Monte Carlo variations. Consequently, while the simulated performance is indicative of design trends, it may not accurately reflect post-fabrication results in a real-world foundry process.

Cadence Virtuoso, while highly capable for schematic and layout-level design, has limitations in frequency-domain simulation resolution and convergence at very high frequencies. During S-parameter simulations, convergence issues were occasionally observed, particularly at fine frequency steps around resonance. These were mitigated using simulation smoothing and mesh tuning, but they still impose a limit on the accuracy of narrowband gain and noise figure estimates.

Moreover, **model parameter tolerances**, such as threshold voltage (Vth), mobility, and capacitance variations, are not fully covered in the default gpdk090 libraries. As a result, temperature dependence and process-voltage-temperature (PVT) corners were not exhaustively explored, which introduces **uncertainty in performance margins** such as gain flatness, noise stability, and power consumption across operating conditions.

Manual biasing design also introduces room for error due to lack of automation or adaptive bias circuits, making it sensitive to transistor mismatch or thermal drift. The simulation assumed ideal passive components, ignoring practical limitations such as inductor Q-factor degradation or parasitic coupling, which could significantly affect high-frequency response in physical implementation.

Despite these limitations, the tools were effectively used to analyze the system under ideal conditions, extract key performance metrics, and validate core design principles. Acknowledging these constraints fosters a more realistic understanding of the gap between simulated and fabricated performance, an essential aspect of practical RF circuit design.

5.5 Impact Assessment (PO(f))

The design and simulation of the 1.8 GHz low-noise amplifier (LNA) extend beyond technical functionality to encompass several critical societal and professional responsibilities inherent to engineering practice. As a core component in wireless communication systems, the LNA plays a vital role in ensuring efficient, reliable signal reception across various applications—ranging from mobile communication and satellite systems to public emergency networks and healthcare monitoring devices.

From a **societal perspective**, improved LNA performance contributes directly to the quality and accessibility of communication infrastructure. In low-resource or rural regions, efficient RF front-end design can enable wider network coverage with lower power requirements, supporting digital inclusion and bridging communication gaps. This aligns with broader societal goals of equitable technology access and economic development through improved connectivity.

In terms of **public health and safety**, wireless systems that rely on robust signal reception—such as remote patient monitoring, wearable medical devices, or emergency alert systems—depend on reliable and noise-tolerant RF circuits. By minimizing noise and maximizing sensitivity, this LNA design enhances the operational reliability of such critical systems, thereby contributing to health and safety indirectly but significantly.

Legal and regulatory compliance in RF design is an implicit responsibility. Although this academic project does not involve physical transmission, real-world deployment of such a design must adhere to international standards (e.g., FCC, ITU) regarding electromagnetic interference (EMI), spectral efficiency, and safe power levels. Understanding these constraints is integral to professional practice, ensuring that engineering solutions are not only effective but also lawful and non-disruptive.

Culturally, the expansion of wireless technology must respect diverse user needs, including language accessibility, data privacy expectations, and technology acceptance across varying demographics. Low-power, cost-effective designs like this one enable widespread adoption of RF technologies without imposing significant financial or environmental burdens—supporting culturally inclusive innovation.

In conclusion, this project acknowledges and integrates societal, health, safety, legal, and cultural considerations into its design philosophy, reflecting the broader ethical and professional responsibilities that engineers must uphold in solving complex, real-world problems.

5.5.1 Assessment of Societal and Cultural Issues

The LNA designed in this project plays a foundational role in wireless communication systems, which are essential to modern society. In developing regions, efficient low-noise amplifiers enable better network coverage with fewer base stations, thereby lowering infrastructure costs and improving access to mobile and internet connectivity. This supports **digital inclusion** and **social development**. From a cultural standpoint, the widespread adoption of wireless technology must respect regional differences in technology literacy, language, and usage patterns. The energy-efficient and cost-effective nature of this design makes it well-suited for integration into diverse social contexts without imposing significant technological or financial barriers

5.5.2 Assessment of Health and Safety Issues

Although the project is simulation-based and does not directly involve physical RF transmission, its real-world application in communication devices entails important health and safety considerations. RF circuits, including LNAs, must operate within regulated power levels to **prevent harmful radiation exposure**. Furthermore, these amplifiers are often embedded in **life-critical systems**, such as medical telemetry, emergency communication devices, and disaster alert systems, where signal integrity is crucial. Ensuring low noise and high sensitivity improves the **safety and reliability** of such systems. Designing for low power also reduces thermal risks and enhances device longevity, contributing to overall user safety.

5.5.3 Assessment of Legal Issues

In practical deployment, all RF circuits must comply with national and international **regulatory standards**, such as those established by the **FCC**, **ETSI**, or **ITU**. These regulations govern emission limits, electromagnetic interference (EMI), frequency allocations, and power levels. Although the gpdk090 PDK used in this project is academic and not intended for fabrication, the design methodology reflects industry standards and encourages **legal awareness** in RF system development. Moreover, future implementation must consider **intellectual property rights**, as component designs and topologies may be subject to patent protections in the RF domain.

5.6 Sustainability and Environmental Impact Evaluation (PO(g))

The design of the 1.8 GHz low-noise amplifier (LNA) reflects an awareness of sustainability principles and responsible engineering practices in both societal and environmental contexts. The project emphasizes **low power consumption**, achieving only 73 mW of simulated power usage. This is especially relevant in today's world where billions of RF front-end circuits operate simultaneously in mobile phones, IoT devices, and sensor networks. Reducing power consumption at the circuit level scales to significant global energy savings, contributing to environmental sustainability and reducing carbon footprints.

From a **societal perspective**, the advancement of efficient RF circuits enables broader deployment of wireless technologies in underdeveloped and remote regions. The low-cost, CMOS-based implementation makes the design viable for mass production, supporting **inclusive access to communication infrastructure**. Improved connectivity enhances opportunities in education, healthcare, agriculture, and commerce—paving the way for social development and economic empowerment.

Furthermore, the project promotes **design practices that consider longevity and reliability**, reducing electronic waste by extending the useful life of RF components. Environmentally, the design avoids unnecessary complexity and minimizes component count, which aligns with eco-conscious manufacturing goals such as reduced material usage and streamlined PCB footprint.

In professional engineering practice, such considerations are vital not only for compliance but also for contributing to long-term sustainability goals. The outcomes of this project demonstrate how responsible circuit design can support **technological advancement** while remaining sensitive to

energy efficiency, **accessibility**, and **environmental stewardship**.

5.7 Ethical Issued (PO(h))

Throughout the execution of this project, ethical principles and professional responsibilities were consciously applied to ensure integrity, transparency, and accountability in all phases of design and reporting. As future engineers, we acknowledged that technical competence must be accompanied by a strong sense of ethical responsibility, especially when working on systems that could ultimately be integrated into critical communication or healthcare infrastructure.

One of the primary ways ethical conduct was maintained was through **academic honesty**. All simulation results, performance metrics, and design choices were presented truthfully without fabrication or exaggeration. Where assumptions or limitations existed—such as using the academic **gpdk090 PDK** instead of an industrial process—we explicitly stated them to preserve the transparency of our work. This reflects our commitment to **honest representation of data and limitations**, a key tenet of engineering ethics.

During literature review and circuit selection, **intellectual property considerations** were respected. We ensured that topologies, techniques, and design ideas sourced from textbooks, research articles, and online repositories were properly credited, avoiding plagiarism. Reusing schematic ideas without attribution would have constituted an ethical violation, which we actively avoided through proper citation and acknowledgment.

An ethical challenge we encountered involved **balancing performance with practicality**. It was tempting to aggressively optimize for gain or noise by violating realistic constraints such as power consumption, matching bandwidth, or transistor reliability. However, we chose to **adhere to realistic design boundaries**, reflecting our commitment to responsible engineering that does not prioritize performance at the cost of sustainability or feasibility.

We also adhered to **collaborative ethics**—maintaining fairness in workload distribution, communication, and decision-making within the project group. Every team member was encouraged to contribute, and disagreements were resolved through mutual respect and technical discussion rather than unilateral decisions.

By internalizing professional norms and ethical practices throughout the project, we have ensured that our work not only meets academic and technical standards but also aligns with the broader expectations of **engineering integrity**, **social responsibility**, and **professional conduct**.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Student ID	Contribution

2006073	Conducted background study and literature review. Performed theoretical matching calculations and output matching using Lout and Cout. Analyzed the effect of Vbias, Lin, Cin, and MOSFET sizing on power and performance. Explored biasing techniques to optimize noise figure (NF), input/output matching (S_{11}, S_{22}), and forward gain (S_{21}).
2006101	Participated in background study and literature review. Contributed to theoretical design using gain, NF, and S_{11} formulas. Focused on power consumption reduction, performed output matching using ADS, and assisted in schematic design and design comparison.
2006158	Worked on background study and topology selection. Led full circuit design including impedance matching and parameter tuning. Provided theoretical justification, implemented cascode configuration, and analyzed trade-offs between NF, S_{11}, and S_{21}. Applied necessary design changes based on simulation results.
2006166	Participated in background study and literature review. Contributed to the design of matching networks involving L_1 and Cin. Conducted schematic simulations to evaluate the effect of Ls, L_1, and transistor width (W) on amplifier performance, and suggested corresponding improvements.

6.2 Mode of TeamWork

The team followed a collaborative and task-oriented approach throughout the project. Responsibilities were divided based on each member's strengths and technical expertise, ensuring efficient progress across all phases. Regular weekly meetings were held to review progress, share findings, and resolve any design challenges. Google Drive and version-controlled folders were used to maintain updated documents, simulation files, and reports. Open communication was encouraged via group chats and online calls, which helped in coordinating schematic development, simulation runs, literature compilation, and result analysis. Decision-making was done collectively, with each member contributing to brainstorming sessions, verifying calculations, and validating simulation results. This distributed yet integrated workflow allowed the team to achieve consistent progress and maintain high-quality output in both documentation and circuit design.

6.3 Log Book of Project Implementation

Week	Milestone Achieved
Week 8	Completed background study and finalized literature review on LNA topologies and RF performance metrics.
Week 9	Derived theoretical equations for impedance matching, gain, and noise figure. Began hand calculations and parameter estimations.
Week 10	Started schematic-level simulation. Determined initial Wopt, L1, L3 values. Designed input matching network.
Week 11	Conducted parametric sweeps to optimize S11 and gain. Fine-tuned biasing conditions (VCE and R).
Week 12	Completed full circuit design including cascode stage. Verified trade-offs among NF, S11, S21. Finalized all passive component values.
Week 13	Performed output matching and full

	simulation. Finalized schematic and simulation model.
Week 14	Prepare the documentation of results. Completed project report and presentation materials.

7 Future Work (PO(l))

- **Lower Overall Power Consumption:**
  Optimize biasing and explore low-power topologies to reduce the 73 mW consumption toward better levels (approximately 20 mW or lower).
- **Redesign Using Active Inductors:**
  Replace bulky passive inductors with MOSFET-based active inductors to reduce area, simplify layout, and improve integration in modern CMOS processes.
- **Layout Design and Parasitic Analysis:**
  Perform layout and extract parasitics to evaluate the real-world impact of capacitances, inductances, and resistances on LNA performance.
- **Linearity Characterization:**
  Measure IIP3 to analyze how well the LNA handles large signals — an essential metric for receiver performance in practical wireless systems.
- **Tape out for Manufacturing:**
  A detailed layout with PCB design for manufacturers to fabricate the hardware and conduct real-time testing with parasitic impedances.

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